Glossary

Absorption	Uptake of moisture by dry foods.
Acid food	A food with a pH of less than 4.6 and a water activity (a_w) equal to or greater than 0.85.
Additives	Chemicals added to food to improve their eating quality or shelf life.
Adiabatic	Changes to the humidity and temperature of air without loss or gain of heat (in drying).
Adiabatic process	Processing in which no heat is added or removed from a system.
Adulterants	Chemicals that are intentionally added to food which are forbidden by law.
Agglomeration	The production of granules from powder particles.
Algorithms	Software building blocks used to construct control sequences in computerised process control.
Alkaline phosphatase	An enzyme in raw milk having a similar *D*-value to heat-resistant pathogens, used to test for effectiveness of pasteurisation.
Annealing	Heating to control the ductility of a material.
Aseptic processing	Heat sterilisation of foods before filling into pre-sterilised (aseptic) containers.
Atomiser	A device to form fine droplets of food (e.g. in a spray drier).
Bacteriocins	Naturally produced peptides that inhibit other micro-organisms, similar in effect to antibiotics.
Baroresistance	Resistance to high pressure.
Barosensitivity	Sensitivity to high pressure.
Biological oxidation demand (BOD)	A measure of the oxygen requirement by micro-organisms when breaking down organic matter, used as a measure of the polluting potential of materials in water.
Black body	A theoretical concept for a material that can either absorb all the heat that lands on it or radiate all of the heat that it contains.
Blancher	Equipment used to blanch foods.
Blanching	Heating foods, especially vegetables, to below 100ºC for a short time, to both inactivate enzymes which would cause a loss of quality during storage and to remove air and soften the food.
Blinding	Blocking of a sieve by food particles.

Bloom	A thin layer of unstable forms of cocoa fat that crystallise at the surface of a coating to produce dullness or white specks.
Botulin	An exotoxin produced by *Cl. Botulinum*, able to cause fatal food poisoning.
Bound moisture	Liquid physically or chemically bound to a solid food matrix which exerts a lower vapour pressure than pure liquid at the same temperature.
Boundary film (or surface film)	Film of fluid next to the surface over which a fluid flows that causes a resistance to heat transfer.
Breading	The application of pre-prepared breadcrumbs to the surface of a food.
Calandria	Heat exchanger used in an evaporator.
Carborundum	An abrasive material made from silicon and carbon.
Case hardening	Formation of a hard impermeable skin on some foods during drying, which reduces the rate of drying and produces a food with a dry surface and a moist interior.
Cashflow	The balance of money at a given time entering and leaving a business.
Cavitation	Production of bubbles in foods by ultrasound and their rapid expansion/contraction.
Centrifugation	The separation of immiscible liquids or solids from liquids by the application of centrifugal force.
Chelating agents	Chemicals which sequester trace metals.
Chemical oxidation demand (COD)	A chemical method used to measure the polluting potential of materials in water.
Chilling	Reduction in the temperature of a food to between $-1^{\circ}C$ and $8^{\circ}C$.
Chilling injury	Physiological changes to some types of fruits and vegetables caused by low temperatures which result in loss of eating quality.
Choke	Restriction of the outlet to a mill to retain particles until sufficiently small (or restriction of the outlet in an extruder).
Climacteric	Abrupt increase in respiration rate in some fruits during ripening.
Clinching	Partial sealing of can lids.
Coating	A generic term to describe the application of a viscous covering (such as batter, chocolate, starch/sugar mixtures) to the surface of a food.
Co-extrusion	The simultaneous extrusion of two or more films to make a co-extruded film or the extrusion of two foods in which a filling is continuously injected into an outer casing in an extruder.
Cold shortening	Undesirable changes to meat caused by cooling before rigor mortis has occurred.
Collapse temperature	The maximum temperature of a frozen food before solute movement causes a collapse of the food structure and prevents movement of water vapour during freeze drying.
Commercial sterility	A term used in heat sterilisation to indicate that processing inactivates substantially all micro-organisms and spores which, if present, would be capable of growing in the food under defined storage conditions.
Common Object Resource Based Architecture (CORBA)	Computer software that acts as an information broker to link process control systems with other computerised company information.
Compound coating	A coating material in which cocoa solids and hardened vegetable oils are used to replace cocoa butter.
Conduction	The movement of heat by direct transfer of molecular energy within solids.
Constant-rate drying	The drying period in which the rate of moisture loss is constant when surface moisture is removed.
Continuous phase	The medium that contains the dispersed phase in an emulsion.

Convection	The transfer of heat in fluids by groups of molecules that move as a result of differences in density or as a result of agitation.
Critical control point (CCP)	A processing factor of which a loss of control would result in an unacceptable food safety or quality risk.
Critical moisture content	The amount of moisture in a food at the end of the constant-rate period of drying.
Crumb	Pre-prepared breadcrumbs used to cover food pieces, or the porous inner part of baked foods.
Crust	Hard surface layer on baked foods.
Cryogen	A refrigerant that absorbs latent heat and changes phase from solid or liquid to a gas, e.g. subliming or evaporating carbon dioxide or liquid nitrogen.
Cryogenic freezers	Equipment that uses subliming or evaporating carbon dioxide or liquid nitrogen directly in contact with food to freeze it.
Cryogenic grinding	Mixing liquid nitrogen or solid carbon dioxide with food to cool it during grinding.
Dead-folding	A crease or fold made in a material that will stay in place.
Decimal reduction time	The time needed to destroy 90% of micro-organisms (to reduce their numbers by a factor of 10).
Depositor	Machine for placing an accurate amount of food onto a conveyor or into a mould.
Desorption	Removal of moisture from a food.
Detergents	Chemicals that reduce the surface tension of water and hence assist in the release of soils from equipment or foods.
Dew point	Temperature at which an air–water vapour mixture becomes saturated with moisture, marking the onset of condensation.
Diafiltration	A process to improve the recovery of solutes by diluting the concentrate during reverse osmosis or ultrafiltration.
Die	A restricted opening at the discharge end of an extruder barrel.
Dielectric constant	The ratio of the capacitance of a food to the capacitance of air or vacuum under the same conditions.
Dielectric heating	A generic term that includes heating by both microwave and radio frequency energy.
Dilatant material	Food in which the consistency increases with shear rate.
Direct heating ovens	Ovens in which products of combustion are in contact with the food.
Dispersed phase	Droplets in an emulsion.
Dosimeter	A device that qualitatively or quantitatively measures the dose of irradiation received by a food.
Dry bulb temperature	Temperature measured by a dry thermometer in an air–water vapour mixture.
Effective freezing time	The time required to lower the temperature of a food from an initial value to a pre-determined final temperature at the thermal centre.
Electrical conductivity	The capacity of a material to conduct electricity.
Electrodialysis	The separation of electrolytes into anions and cations by the application of a direct electrical current and the use of ion-selective membranes.
Emulsification	Creation of an emulsion by the dispersion of one immiscible liquid (dispersed phase) in the form of small droplets in a second immiscible liquid (continuous phase).
Emulsifying agent	Chemical that forms micelles around each droplet in the dispersed phase of an emulsion to reduce interfacial tension and prevent droplets from coalescing.
Enrobing	The unit operation in which food pieces are coated with chocolate or other materials.

Entrainment	Oil droplets that are carried over in steam produced by vigorously frying foods, leading to loss of oil, or loss of concentrated droplets of product with vapour during evaporation by boiling.
Equilibrium moisture content	The moisture content of a food at which it neither gains nor loses moisture to its surroundings (at a given temperature and pressure, the food is in equilibrium with the air vapour mixture surrounding it).
Equilibrium relative humidity	Relative humidity of the storage atmosphere in equilibrium with the moisture content of food.
Eutectic temperature (in freezing)	The temperature at which a crystal of an individual solute exists in equilibrium with the unfrozen liquor and ice.
Exhausting	Removal of air from a container before heat processing.
Expeller	A horizontal barrel, containing a helical screw, used to extract oil from seeds or nuts.
Expression	The separation of liquids from solids by applied pressure.
Extractors	Equipment used to extract food components using solvents.
Extruder	One or more screws rotating in a barrel with restricted apertures at the discharge end, used for producing extruded foods.
Extrusion	A process that involves the combination of several unit operations including mixing, cooking, kneading, shearing, shaping and forming to produce extruded foods.
F-value	The time required to destroy a given percentage of micro-organisms at a specified reference temperature and *z* value.
Falling-rate drying	The drying period in which the rate of moisture loss declines.
Feedback control	Automatic control of a process using information from sensors to adjust processing conditions.
Feed-forward control	Comparison of processing conditions with a model system, used in automatic process control.
Field heat	Heat within crops when they are harvested.
Filter cake	Solids removed by filtration.
Filter medium	Porous material through which food is filtered.
Filtrate	The liquor remaining after solids are removed by filtration.
Filtration	The separation of solids from liquids by passing the mixture through a bed of porous material.
Final eutectic temperature (in freezing)	The lowest eutectic temperature of solutes in equilibrium with unfrozen liquor and ice.
Flash pasteurisation	Heat treatment involving temperature greater than 72ºC for 15 s for milk. (Also known as higher-heat shorter-time processing.)
Flash-over	Arcing of electricity between electrodes without heating taking place.
Fluence	Energy imparted by light to the surface of a material.
Flux	Flow of liquid through reverse osmosis or ultrafiltration membranes.
Foam	A colloidal system with a liquid or solid continuous phase and a gaseous dispersed phase.
Forming	Moulding of doughs and other materials into different shapes.
Fouling	Deposits of food or limescale on surfaces of heat exchangers.
Free moisture	Moisture in excess of the equilibrium moisture content at a particular temperature and humidity, and so free to be removed.
Freeze concentration	Concentration of liquid foods by freezing water to ice and removal of ice crystals.
Freeze drying	Dehydration of food by freezing water to form ice, followed by removal of ice by sublimation.
Freezing plateau	The period during freezing when the temperature of a food remains almost constant as latent heat of crystallisation is removed and ice is formed.

Friability	The hardness of a food and its tendency to crack.
Grading	The assessment of a number of attributes to obtain an indication of overall quality of a food.
Grey body	A concept used to take account of the fact that materials are not perfect absorbers or radiators of heat.
Half-life	The time taken for an isotope to lose half of its radioactivity.
Hazard analysis	The identification of potentially hazardous ingredients, storage conditions, packaging, critical process points and relevant human factors which may affect product safety or quality.
Headspace	The space in a container between the surface of a food and the underside of the lid.
Heat sterilisation	Destruction of the majority of micro-organisms in a food by heating.
Hermetically-sealed container	A package that is designed to be secure against entry of micro-organisms and maintain the commercial sterility of its contents after processing.
Heterofermentative micro-organisms	Micro-organisms that produce more than one main metabolic product.
Homofermentative micro-organisms	Micro-organisms that produce a single main byproduct.
Homogenisation	The reduction in size and increase in number of solid or liquid particles in the dispersed phase.
Humectants	Chemicals (e.g. salt, sugar, glycerol) that are able to lower the water activity in a food by depressing the vapour pressure.
Hydrocooling	Immersion of fruits and vegetables in chilled water.
Hydrophile-lipophile balance (HLB value)	The ratio of hydrophilic to hydrophobic groups on the molecules of an emulsifier.
Hygroscopic foods	Foods in which the partial pressure of water vapour varies with the moisture content.
Hydrostatic head	The pressure resulting from the weight of a column of liquid.
Hyperfiltration	Reverse osmosis.
Impact strength	The force required to penetrate a material.
Indirect heating ovens	Ovens in which heat from combustion is passed through a heat exchanger to heat air which is then in contact with the food.
Inventory	The stored accumulation of materials in an operation.
Ion exchange	The selective removal of charged molecules from a liquid by electrostatic adsorption, followed by their transfer to a second liquid using an ion-exchange material.
Ionisation	Breakage of chemical bonds (e.g. during irradiation).
Irradiation	The use of γ-rays to preserve foods by destruction of micro-organisms or inhibition of biochemical changes.
Isostatic	Uniform pressure throughout a food.
Isotope	A source of γ-rays from a radioactive material such as cobalt-60 or caesium-137.
Just-in-time	Management system in which goods are ordered as they are required and stocks are not held in warehouses.
Kinetic energy	Energy due to motion.
Lamination	Bonding together of two or more packaging films, papers or foods.
Latent heat	Heat taken up or released when a material undergoes a change of state.
Leaching	Washing out of soluble components from the food.
Lethality	Integrated effect of heating temperature and time on micro-organisms.
Loss factor	A measure of the amount of energy that a material will dissipate when subjected to an alternating electric field (in microwave and dielectric heating). (Also termed the ‘dielectric loss’ or ‘loss tangent’.)

Low acid food	A food with a pH greater than 4.6 and a water activity (a_w) equal to or greater than 0.85.
Manufacturing resource planning	Computer-based systems used to control distribution networks by using forecasted demand for and actual orders to assist management decisions.
Material requirement planning	A single integrated computer system, containing a database that can be accessed by all parts of the company for management planning.
Mechanical refrigerators	Equipment which evaporates and compresses a refrigerant in a continuous cycle, using cooled air, cooled liquid or cooled surfaces to freeze foods.
Metallisation	A thin coating of aluminium on plastic packaging.
Microfiltration	A pressure-driven membrane process using membranes with a pore size of 0.2–2 μm at lower pressures than ultrafiltration.
Microwaves	Energy produced commercially at frequencies of 2450 MHz for domestic ovens, 896 MHz for industrial heating in Europe and 915 MHz for industrial heating in the USA.
Mimetics	Low calorie fat substitutes.
Mimic panel	A graphical display of a process.
Moulders	Machines that form dough or confectionery into different shapes.
Multiple effect	The re-use of vapour from boiling liquor in one evaporator as the heating medium in another evaporator operating at a lower pressure.
Nanofiltration	A membrane process to separate particles with molecular weights from 300–1000 Da, using lower pressures than reverse osmosis.
Neural networks	Computer systems that are able to analyse complex relationships in a process and 'learn' from experience.
Nip	The gap between rollers in a mill or a moulding/forming machine.
Nominal freezing time	The time between the surface of the food reaching 0ºC and the thermal centre reaching 10ºC below the temperature of the first ice formation.
Non-hygroscopic foods	Foods that have a constant water vapour pressure at different moisture contents.
Non-Newtonian liquid	Food in which the viscosity changes with rate of shear.
Nucleation	The formation of a nucleus of water molecules that is required for ice crystal formation.
Ohmic heating	Direct electrical heating of foods.
Overall heat transfer coefficient (OHTC)	The sum of the resistances to heat flow due to conduction and convection.
Panning	The process of building up thin layers of sugar, sweetener or other coatings in a controlled way onto solid cores of nuts, fruit, etc.
Pasteurisation	A relatively mild heat treatment in which food is heated to below 100ºC to preserve it without substantial changes to sensory characteristics or nutritional value. In low acid foods, the main reason for pasteurisation is destruction of pathogens.
Pinholes	Small holes in can seams or flexible packaging.
Plasticiser	Chemicals added to plastic films to make them more flexible.
Polymorphic fat	A fat that can crystallise into more than one form.
Potential energy	Energy due to position of an object.
Preforms	Small dense pellets made in an extruder from pre-gelatinised cereal dough, which are suitable for extended storage until they are converted to snackfoods by frying, toasting or puffing. (Also known as 'half-products'.)
Press cake	Solid residue remaining after extraction of liquid component from foods.
Process inter-locking	Linking different parts of a process so that one cannot operate until a

	second is correctly set up.
Programmable logic controllers (PLCs)	A microcomputer that is used in process control to replace electrical relays and to collect and store process data.
Pseudoplastic material	Food in which the viscosity decreases with increasing shear rate.
Psychrometrics	The study of inter-related properties of air–water vapour systems.
Radiation	The transfer of heat by electromagnetic waves.
Radio frequency energy	Energy produced commercially at frequencies of 13.56 MHz, 27.12 MHz or 40.68 MHz for industrial heating.
Radiolysis	Changes to a food material caused by ionising radiation to produce chemicals that destroy micro-organisms, etc.
Recrystallisation	Physical changes to ice crystals (changes in shape, size or orientation) which are an important cause of quality loss in some frozen foods.
Redox potential	Oxidation/reduction potential of a food or microbial substrate.
Refrigerant	A liquid that has a low boiling point and high latent heat of vaporisation so that it can change phase and absorb or lose heat in a refrigerator.
Refrigerators	Equipment that evaporates and compresses a refrigerant in a continuous cycle, using cooled air, cooled liquid or cooled surfaces to freeze foods.
Relative humidity	The ratio of the partial pressure of water vapour in air to the pressure of saturated water vapour at the same temperature, multiplied by 100.
Respiration	Metabolic activity of living animal or plant tissues.
Retort	A pressurised vessel used to heat foods above 100ºC during canning.
Reverse osmosis	Unit operation in which small molecular weight solutes (with molecular weights of approx. 100 DA) are selectively removed by a semi-permeable membrane under high pressure.
Screen	A sieve.
Sensible heat	Heat used to raise the temperature of a food or removed during cooling, without a change in phase.
Sequence control	A type of process control in which the completion of one operation signals the start of the next.
Soils	A generic term used for all types of contaminating materials on foods or equipment.
Sorption isotherm	A curve produced from different values of relative humidity plotted against equilibrium moisture content.
Sorting	The separation of foods into categories on the basis of a measurable physical property.
Specific electrical resistance	Electrical resistance of a food between two 1 cm^2 electrodes that are located 1 cm apart (i.e. the resistance of 1 cm^3 of product), having units of ohms cm^{-2} cm^{-1}.
Specific growth rate	The slope of the curve when the natural logarithm of microbial cell concentration is plotted against time.
Specific heat	The amount of heat that accompanies a unit change in temperature by a unit mass of material.
Stabilisers	Hydrocolloids that dissolve in water to form viscous solutions or gels.
Steady-state heat transfer	Heating or cooling when there is no change in temperature at any specific location.
Sterilants	Chemicals that inactivate micro-organisms.
Streamline (or laminar) flow	Flow of liquids in layers without significant mixing between layers.
Sublimation	A change in state of water directly from ice to water vapour without melting.
Substrate	A growth medium for micro-organisms.

Supercooling	A phenomenon in which water remains liquid although the temperature is below its freezing point.
Supercritical carbon dioxide	Liquid CO_2 used to extract food components.
Supervisory Control and Data Acquisition (SCADA)	A type of computer software that collects data from programmable logic controllers and displays it as graphics to operators in real-time.
Surface heat transfer coefficient	A measure of the resistance to heat flow caused by a boundary film of liquid.
Susceptor	A packaging material that is used to create a localised high temperature in microwave ovens; usually made from lightly metallised polyethylene terephthalate.
Tempering	Cooling food to close to its freezing point, or a process of re-heating, stirring and cooling chocolate to remove unstable forms of polymorphic fats.
Tensile elongation	A measure of the ability to stretch.
Tensile strength	The force needed to stretch a material.
Thermal centre	The point in a food that heats or cools most slowly.
Thermal conductivity	A measure of the heat transfer properties of solid materials.
Thermal death time (TDT) or *F*-value	The time required to achieve a specified reduction in microbial numbers at a given temperature.
Thermal diffusivity	The ratio of thermal conductivity of a product to specific heat, multiplied by the density.
Thermal shock	Heating: fracture to a glass container caused by rapid changes in temperature; freezing: a rapid reduction in temperature that causes foods to fracture.
Ultra high temperature (UHT)	Processing heat sterilisation at above 135ºC for a few seconds.
Ultrafiltration	Unit operation in which solutes having molecular weights in the range of 1–200 kDA are selectively removed using a semi-permeable membrane operating at lower pressure than reverse osmosis.
Ultrasonication	Treatment of foods using ultrasound.
Unitised loads	Grouping of packages into larger loads.
Usage value	The rate of usage of individual materials in an inventory multiplied by their individual value.
Unsteady-state heat transfer	Heating or cooling where the temperature of the food and/or the heating or cooling medium are constantly changing.
Venting	Removal of air from a retort before heat processing.
Viscoelastic material	Food materials which exhibit viscous and elastic properties including stress relaxation, creep and recoil.
Voidage	The fraction of the total volume occupied by air (the degree of openness) of a bed of material in fluidised-bed drying.
Water activity	The ratio of vapour pressure of water in a solid to that of pure water at the same temperature.
Web	A packaging film.
Wet bulb temperature	Temperature measured by a wet thermometer in an air–water vapour mixture.
Yield	Weight of food after processing compared to weight before processing.
Young's modulus	(also modulus of elasticity) = stress/strain and is a measure of the hardness of a material.